

MAGNETOM

Installation Instructions System

Third Party Helium Filling

Aera
Skyra
Spectra

Document Version

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1.1 General

The 4K magnet design does not include a permanently fitted syphon. A short magnet service syphon must be fitted when filling with liquid helium. The service syphon is tooling only and must be removed immediately after filling has been completed.

This procedure can be performed on magnets at field and magnets at zero field.



To recover a warm magnet (Post quench) refer to magnet temperatures in the **Bottom filling procedure** (→ Filling the Magnet - Bottom Fill / Page 43)

This document describes the filling procedure for the 4K magnet systems:

- OR98 - Area.
- OR99 - Skyra.
- OR103 Skyra and Spectra.

These magnets are fitted with the 2306 MSUP (Magnet supervisory unit). The MSUP provides several options for monitoring helium levels during filling.

These options are:

- Connection of a laptop PC to the RS232.
- Direct from the LEDs on the MSUP front cover.

1.2 Safety



Check that the Quench line installation is complete and the magnet connected before helium filling.



WARNING

Strong magnetic field.

Failure to observe the following safety warning may result in physical injury and equipment damage.

- » Only non-magnetic equipment may be taken into the magnet room.



DANGER

Risk of asphyxiation.

Failure to observe the following safety warning may result in dizziness and loss of consciousness.

- » Oxygen monitors must be activated at all times.
- » Do not vent helium gas continuously into the magnet room during helium transfer.
- » Check room ventilation is turned on.



CAUTION

Cryogenic liquids.

Failure to observe the following safety warning may result in cold burns.

- » Protective clothing, cryogenic gloves and eye protection must be worn.
- » Filling must be performed only by trained personnel.



CAUTION

Pressurized helium vessel.

Failure to observe the following safety information may result in physical injury.

- » Do not disconnect any of the turret venting or fittings without first depressurizing the helium vessel.

**CAUTION**

Risk of quench during helium filling

If the magnet quenches:

- » Exit room and wait 15 to 20 minutes for magnet pressure to fall.
- » Stop the helium transfer as soon as it is safe.
- » Follow Post Quench procedures.
- » Report quench and fill in quench database.

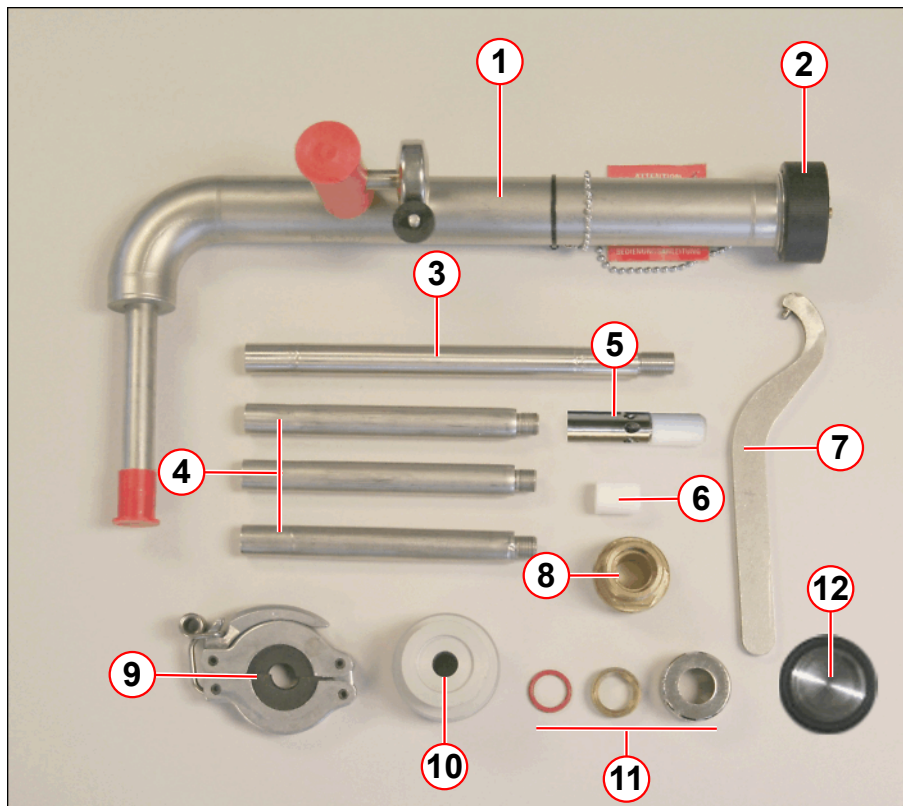
NOTICE

Mishandling of the Dewar.

- » The safety regulations of the dewar manufacturer must be observed.

1.3 Equipment and Tools

Fig. 1: Magnet service syphon kit



- (1) Service syphon assembly
- (2) Syphon end cap
- (3) 200 mm extension leg (1 off)
- (4) 140 mm extension legs (3 off)
- (5) 6 hole helium fill diffuser tip (top fill)
- (6) 22mm PTFE seal (bottom fill)
- (7) Syphon C-spanner/wrench
- (8) M28 brass syphon adaptor bush (OR105, OR122)
- (9) Syphon leg support clamp (low ceiling)
- (10) Nitrogen fill adaptor and O-ring seal
- (11) Syphon O-ring seal, brass syphon flange, syphon ring nut
- (12) NW25 flow restrictor

- Magnet service syphon kit (material number **10126651**).
- Dewar syphon.
- Pressure control service tool (material number **03861221**).
- Dewar neck adaptor, clamps, seals, and hose.
- Heat gun & paper toweling.
- Helium gas (99.996% - 4.6 purity).
- Pressure regulator 200 bar/ 0 - 1.5 bar.
- Protective clothing: gloves, safety glasses, face shield, lab coat, safety shoes.
- Replacement copper seal for syphon port on the turret (material number **03108094**).

- Gas bottle wrench.
- A room or personal O₂ Monitor.
- Edwards hand-held leak detector - (material number **10614850**).
- Service platform.
- Service tool kit.
- Laptop computer and RS232 / USB cable.

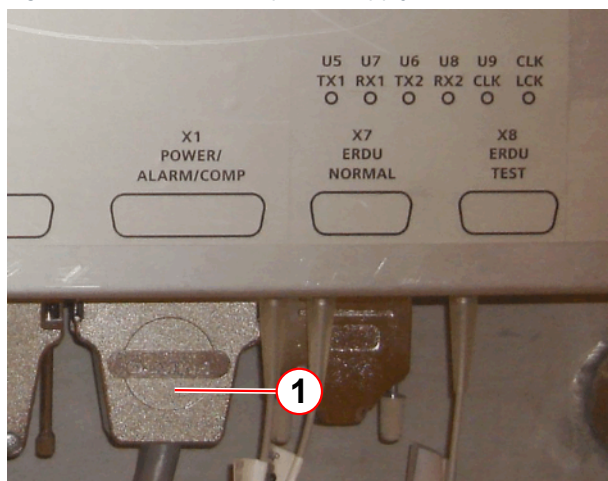
1.3.1 External 36 volt power supply

If there is no mains power to the MSUP additional Service tooling is available:

- External 36-V PSU kit (material number **08396371**) to provide power to the MSUP.
- External 36-V PSU MSUP cable (material number **10612362**).

With the 36 Volt providing power to the 2306 MSUP, the magnet pressure heaters will be activated when the magnet pressure drops to below 0.2 psi (G). This function will occur when the magnet liquid helium vessel is depressurized to allow fitting of the Service syphon.

Fig. 2: Connect the 36V power supply at X1



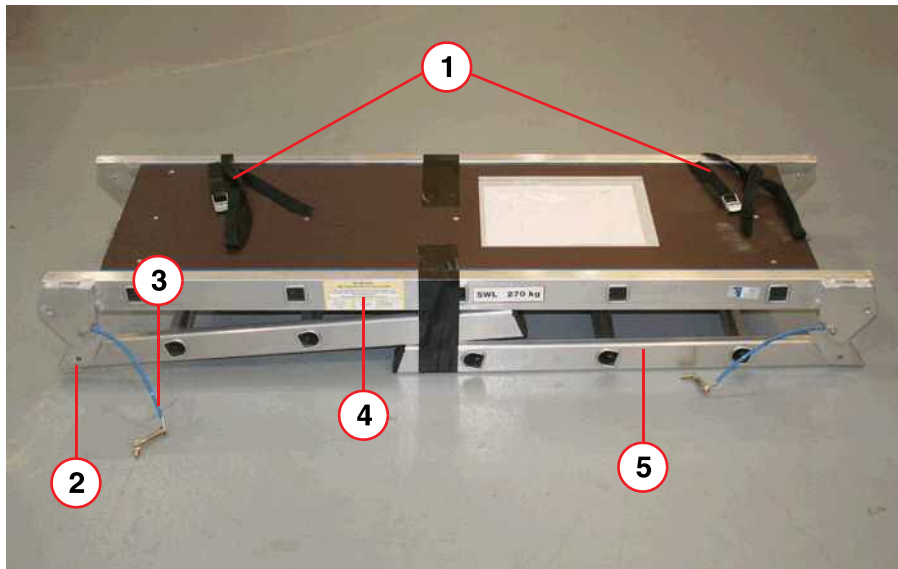
(1) Connector X1

To enable magnet readings to be taken from the MSUP, connect an external 36v power supply.

- Connect the 36v PSU kit
 - Connect the mains lead between the power supply and local power source.
 - Connect the adaptor lead between X1 MSUP (D type connector) and the power supply.
 - Turn ON the 36v mains power source.
- Check the helium level by pressing the MSUP SAMPLE switch, **OR**:
 - Check the helium level from a service laptop.

1.4 Service Platform Installation

Fig. 3: Service platform - as delivered



- (1) Safety straps (2 off)
- (2) Hinged locking pin location (each end)
- (3) Retaining wire and hinged locking pin (each end)
- (4) Warning label
- (5) Step legs (each end)

1.4.1 Service platform safety

Before using the service platform **read the following safety cautions** and perform the **required safety checks**.



CAUTION

Risk of personal injury

Failure to observe the following precautions may result in personal injury.

- » Always check the platform and hinge locking pins for damage.
- » Always check the hinge locking pins are secured.
- » Always secure the platform to the magnet frame or with the two long straps through the magnet bore.
- » Always ascend and descend (Walk UP or DOWN) the steps facing the platform.
- » Where possible use the frame or covers for additional hand support.
- » Never climb onto the platform from the side.
- » Never jump from the steps or top of the platform.

Fig. 4: Correct method - Up to platform



Fig. 5: Incorrect method - Down from platform

**⚠ CAUTION****Risk of personal injury**

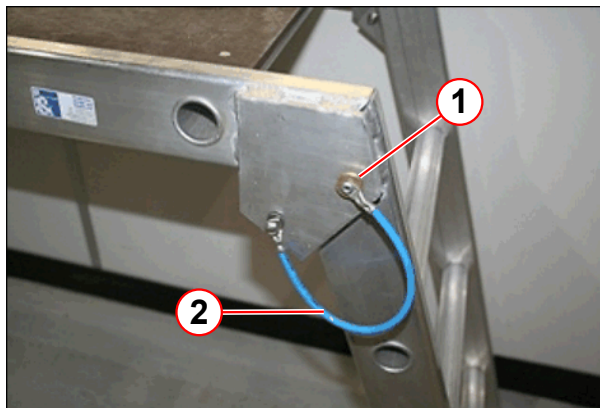
Carrying heavy components or tooling up or down the platform steps can lead to injury.

- » DO NOT carry heavy components up or down the platform steps.
- » Place heavy objects (i.e. cold head) onto the platform first.
- » Only lift the component when standing safely on the platform.

1.4.2 Platform assembly

- Remove any packaging.
- Unfold both step legs.

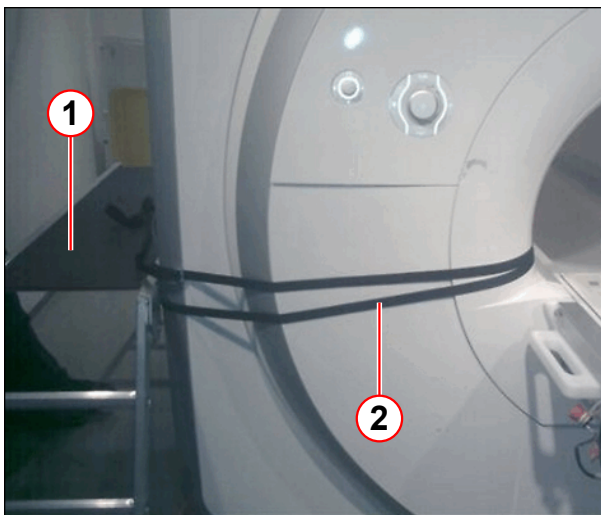
Fig. 6: Service platform - lock legs in position



- (1) Hinged locking pin
(2) Retaining wire

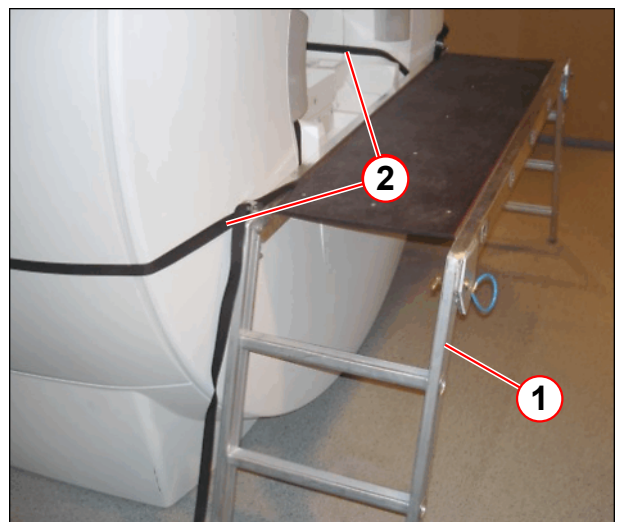
- Align the hole at the top of the leg with the **pin location hole** (pos 1).
- Push the **hinged locking pin** through the hole (pos 1) and fold back the hinge to secure.
- Repeat with the other leg.

Fig. 7: Securing the service platform (example)



- (1) Service platform
(2) Long straps (fitted through the magnet bore)

Fig. 8: Securing the service platform (example)



- (1) Service platform
(2) Long straps (fitted through the magnet bore)



Do not exceed the maximum load - the combined weight of 2 people must not exceed 270 kg.



The fiber optic connections in the top of RFIS are very fragile, so exercise due caution.

1.5 Preparation

Site preparation

- Ensure warning signs are set up and visible.
- Ensure there is a clear path to exit the magnet room or fill area quickly.

System preparation



CAUTION

Cold head failure.

The cold head seals can fail, and leak to atmosphere.

- » Always ensure the cold head bypass valve is **CLOSED** during helium filling.

- Remove the appropriate system covers to access the magnet turret and MSUP.
- Position the Service platform next to the magnet and secure using the 2 x straps.
- Check that there is **open and clear access** to the magnet for helium dewar(s).
- Check that there is **open and clear access** for fitting the syphon.

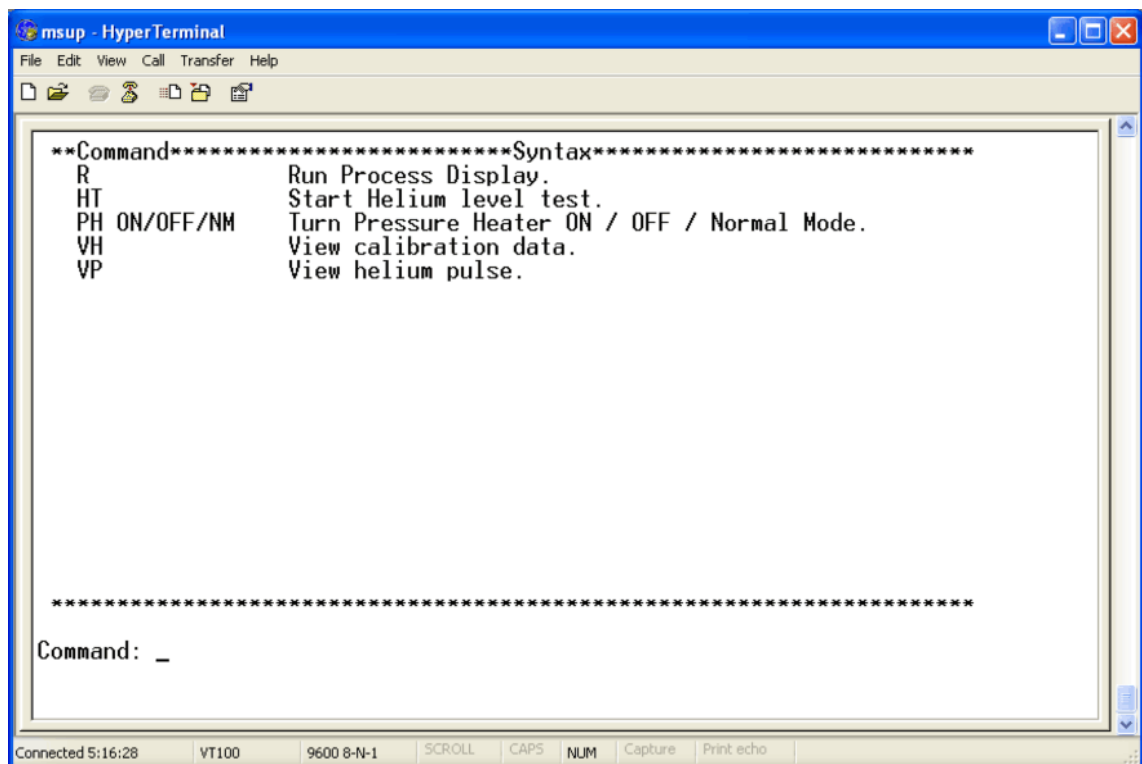
1.6 Third Party Laptop Connection



Third party companies have limited access to the MSUP command screens.

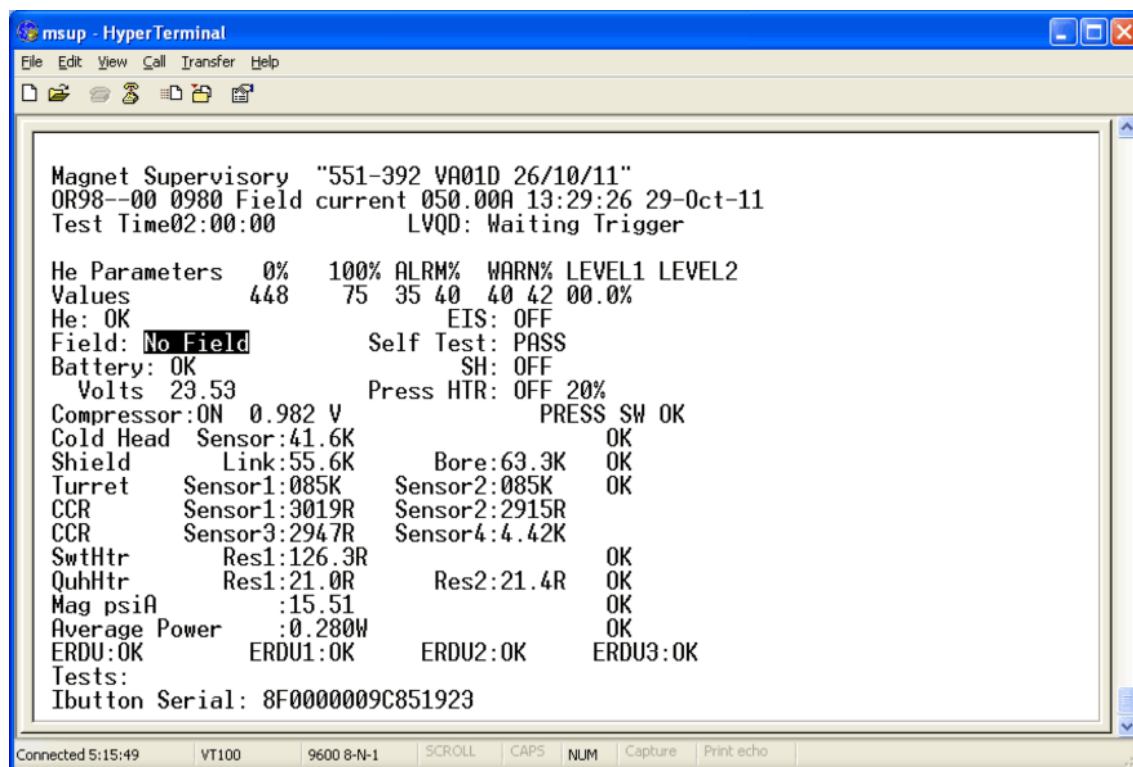
- A service laptop can be connected to the MSUP. This will allow access to the Service software to enable pressure control and accurate helium levels:
- Turn on the laptop and open a Terminal Emulation application (for example HyperTerminal).
 - Set bits per second to 9600.
 - Set data bits to 8.
 - Set parity to 1.
 - Set flow control to none.
 - Set emulation to VT100 (File: Properties: Settings).
- Connect the laptop to connector **X6** on the MSUP via an RS232 cable.
- Press <ESC> to display the MSUP Option screen.

Fig. 9: Option screen (for 3rd Party companies)



- Press <R> enter to display the MSUP Process display screen.

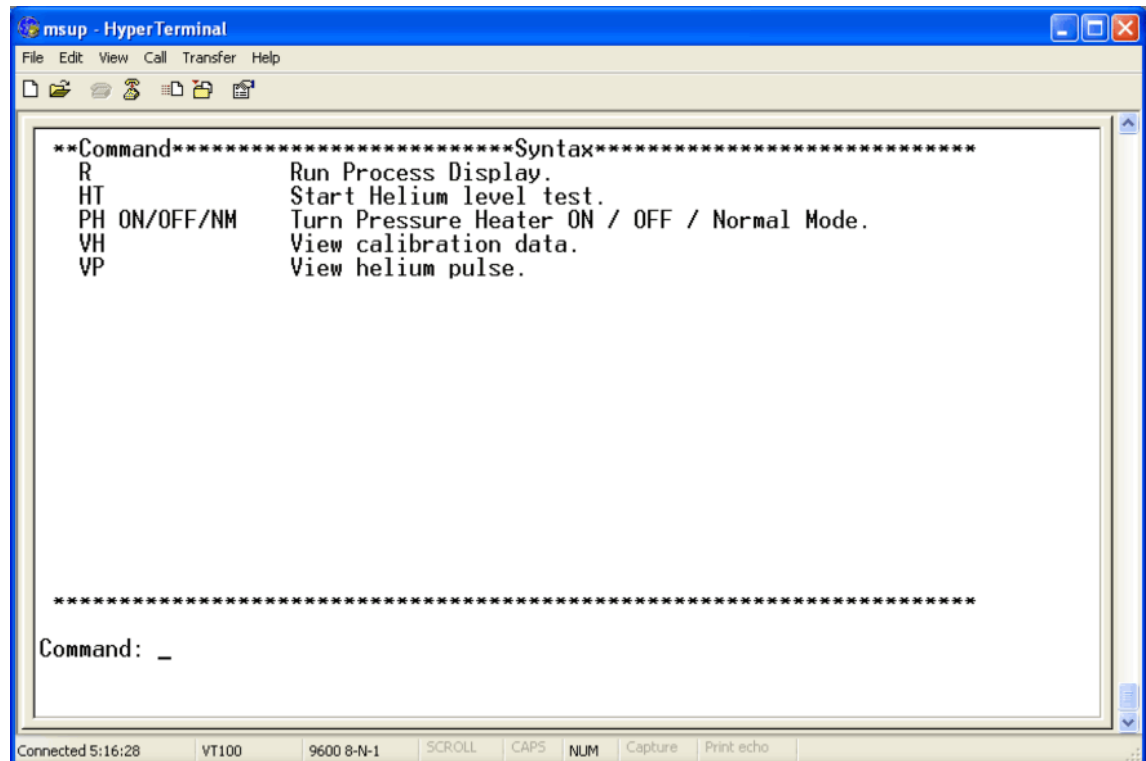
Fig. 10: MSUP Process display screen (Example)



1.6.1 Perform a Helium Sample

- Press <ESC> to display the **MSUP** option window (for third party companies).

Fig. 11: Option screen (for 3rd Party companies)



- Press **HT** <ENTER>. Wait until the sample is completed.
- Press **R** <ENTER> to display the new helium value on the MSUP Process display screen.

Tab. 1 MSUP commands

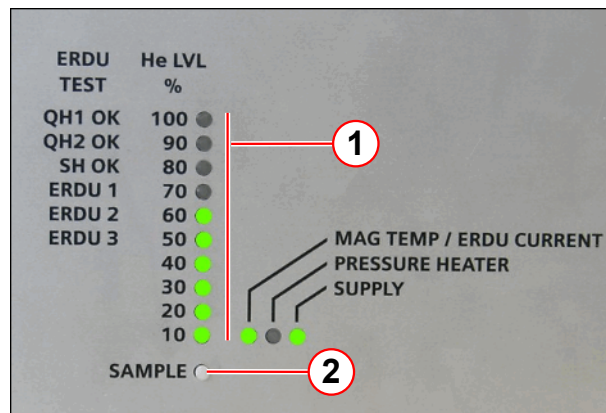
MSUP RS232 command, followed by <ENTER>	System operation
HT	A new helium sample
PH ON	Turns ON the pressure heater
PH OFF	Turns OFF the pressure heater
PH NM	Turns the pressure heater to the normal running mode
VH	Displays the helium calibration parameters
VP	Displays the helium pulse information
R	Displays the process image



After every helium fill the LVQD function must be reset by re-entering the command '**RD**' **<ENTER>**.

1.7 Helium Level Check

Fig. 12: MSUP 2306 - Helium level sample (transit mode)



- (1) Helium level % scale
(2) Sample switch

- Locate the **SAMPLE** switch on the MSUP lid.
- Depress the sample switch for 2 s, and release.
- Wait until the sample is completed, (lights remain static).
- The **green LEDs** illuminated, indicate the actual liquid helium level.

Determine the type of helium fill required:-

- Top fill with 10% helium and temperature LED Green, Normal).
- Bottom fill with zero helium and temperature LED Yellow.

Tab. 2 MSUP 2306 - Magnet temperature indicators

Temperature	LED status	Action	
Room to nitrogen (77K)	RED	Check the magnet CCR temperatures	Contact magnet HSC
Nitrogen to helium	YELLOW	Refill immediately	Bottom fill
Helium temperature normal (4.2K)	GREEN	Refill as required	Top fill

1.8 Magnet Depressurization

To reduce helium boiloff ensure that the pressure heater is turned off. Deactivation of the pressure heater is achieved by one of the following options:

1. From the MSUP option screen type **PH OFF**, to turn the pressure heater OFF. The pressure heater will automatically activate after 60 minutes.
2. Fit the pressure service tool (**03861221**). The tooling forces a reading of 18.5 psi (A) within the MSUP. When fitted the pressure heater will not turn on.
3. Turn off the MSUP. This will ensure that no power is put into the pressure heater.



Check the magnet gauge pressure before opening the bypass vent valve. This pressure value is required later when setting the transfer pressure from the helium dewar.

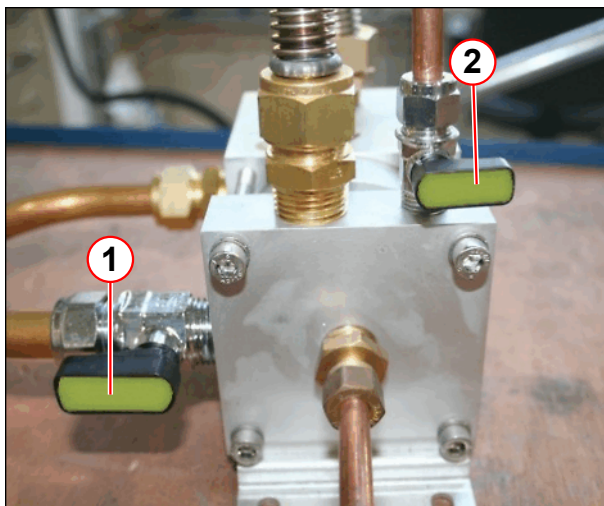
Depressurizing the magnet will take:

- 20 to 30 minutes with a magnet site at sea level.
- Up to 2 hours depending on the altitude of the magnet site above sea level.
- Use the depressurization time to prepare:
 - » The helium dewar.
 - » The dewar and service syphons.

To reduce helium boiloff ensure that the pressure heater is turned off.

- From the option screen type **PH OFF**, to turn the pressure heater OFF.

Fig. 13: Bypass valve positions for depressurization



- (1) Magnet bypass valve OPEN
(2) Cold head bypass valve CLOSED

To depressurize the magnet:

- Fully open the magnet bypass valve.
- At **0.2 psi (G)** above atmospheric pressure the compressor will stop.
- Wait until **0.1 psi (G) pressure** is displayed on the pressure gauge.
- Warm up the vent valve assembly using a heat gun.
 - » Take care not to damage any components when using a heat gun.
- **Close** the magnet bypass valve.

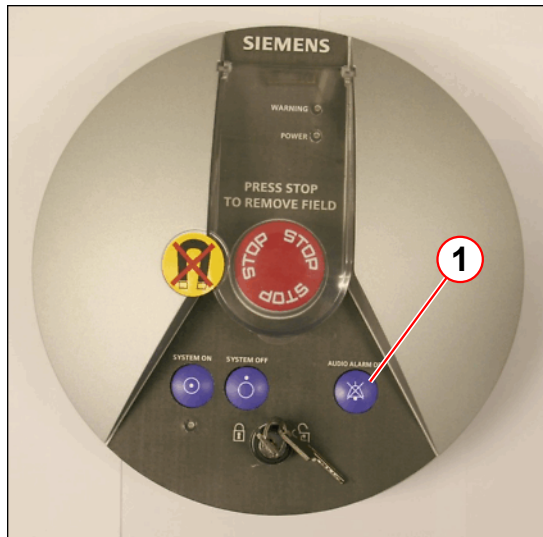
**CAUTION**

Cold head failure.

The cold head seals can fail, and leak to atmosphere.

- » Always ensure that the cold head bypass valve is **CLOSED** during helium filling.

Fig. 14: Alarm box front cover



(1) Audio alarm OFF button

- The alarm box will sound when pressure falls to alarm limits.
 - » The yellow **WARNING LED** lights on the alarm box.
 - » The alarm buzzer is activated
 - » Press **AUDIO ALARM OFF** to silence the alarm.

1.9 Helium Dewar Preparation

This procedure describes the use of **top fill** helium dewars used with the Siemens Magnet Technology dewar syphon.

The procedure covers all dewars that are externally pressurized with helium gas.

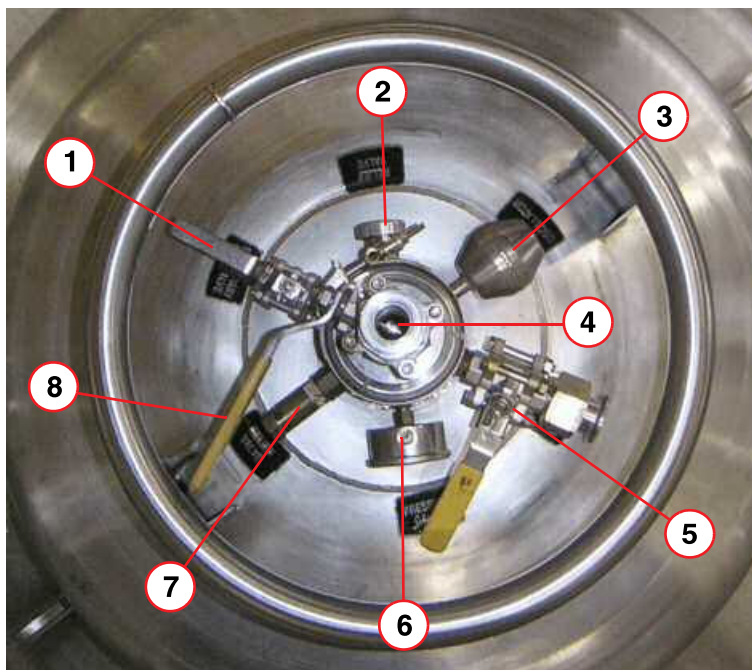


Refer to the dewar manufacturer's documentation when using side fill and self-pressurizing Dewar vessels. Self-pressurizing Dewar vessels only deliver 4 & 8 psi when the fill is performed at sea level. This may not be the most efficient pressure at higher altitudes.



Refer to the dewar manufacturer's Safety Information and check the Dewar valve functions before commencing the helium fill.

Fig. 15: Dewar Ports (typical)

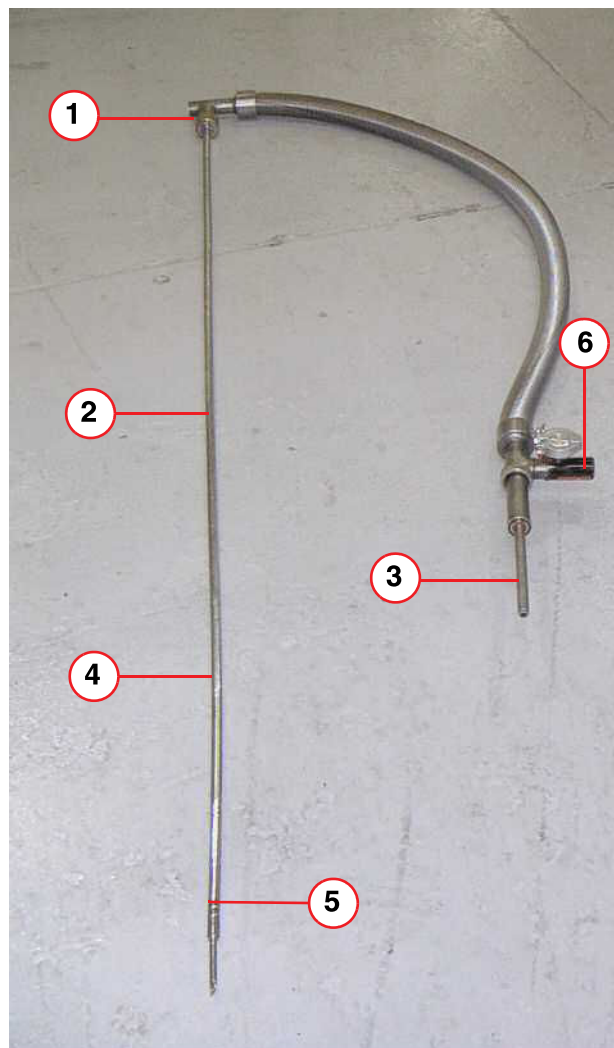


- (1) Low pressure relief valve (shown open)
 - (2) Relief valve
 - (3) Anti-oscillator
 - (4) Syphon entry port (shown closed)
 - (5) Primary ball valve (shown closed)
 - (6) Pressure gauge
 - (7) Relief valve
 - (8) Syphon entry valve
- If the helium dewar vessel has high pressure (greater than 0.3 bar (4.4 psi (G))):
 - **Slowly open** the primary ball valve to release the dewar pressure.
 - Depressurization must be done in a **well ventilated area - NOT in a building**.
 - Check for the cause of the high pressure. Inform the supplier if there are faulty dewar valves.
 - Fit the dewar syphon adaptor to the top of the dewar.

- **Close** the primary ball valve and low pressure relief valve.

1.10 Dewar Syphon Preparation

Fig. 16: Dewar syphon



- (1) Syphon flow valve
- (2) Syphon leg
- (3) Bayonet connection (male)
- (4) Extension leg
- (5) Sludge inhibitor
- (6) Syphon vacuum valve

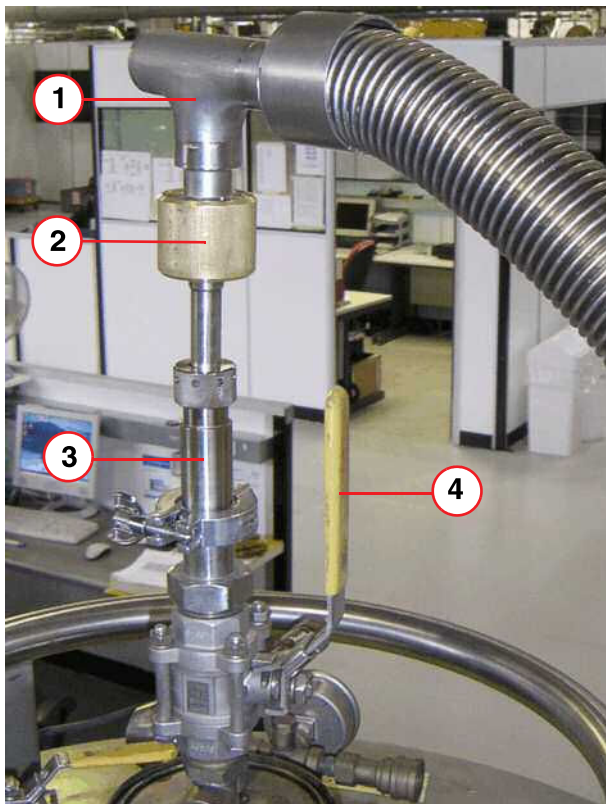
- Fit the extension leg (s) and sludge inhibitor, to the dewar syphon leg.
- The helium dewar must be positioned close to the magnet with the dewar syphon fitted.
- There must be no excessive bending of the syphon.



Do not open the syphon vacuum valve (6).

1.11 Fitting the Dewar Syphon

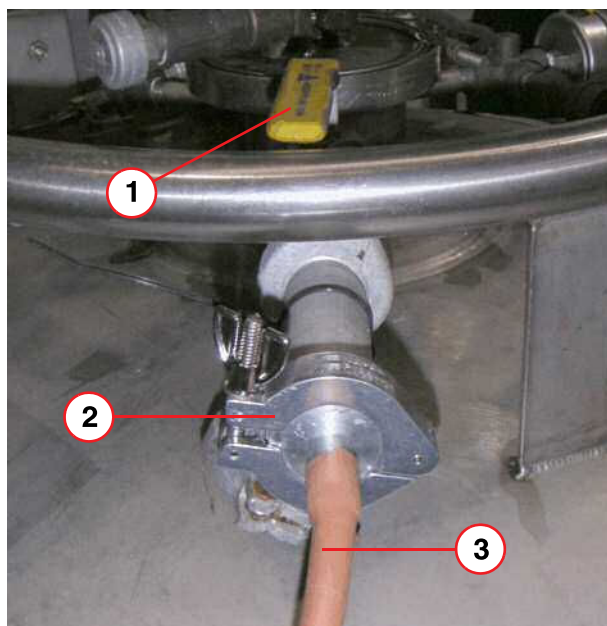
Fig. 17: Dewar syphon fitment



- (1) Syphon
- (2) Syphon flow valve
- (3) Dewar syphon adaptor
- (4) Syphon entry valve

- **Open the syphon flow valve** (pos 2) two full turns.
- Insert the syphon leg into the **dewar syphon adaptor** (pos 3), and tighten the ring nut and seal in order to create a gas seal that the syphon can slide through.
(Do not fully tighten the ring nut at this stage).
- **Open the syphon entry valve** (pos 4), and slowly slide the syphon into the dewar. Gas will begin to flow through the syphon.
- When the sludge inhibitor touches the bottom, **fully** tighten the ring nut and seal.
- The syphon will now be purged. The syphon flow valve (pos.2) can be **closed**.
- Open the dewar relief valve if required to keep pressure at **4–5 psi (G)**.

Fig. 18: Dewar gas connection

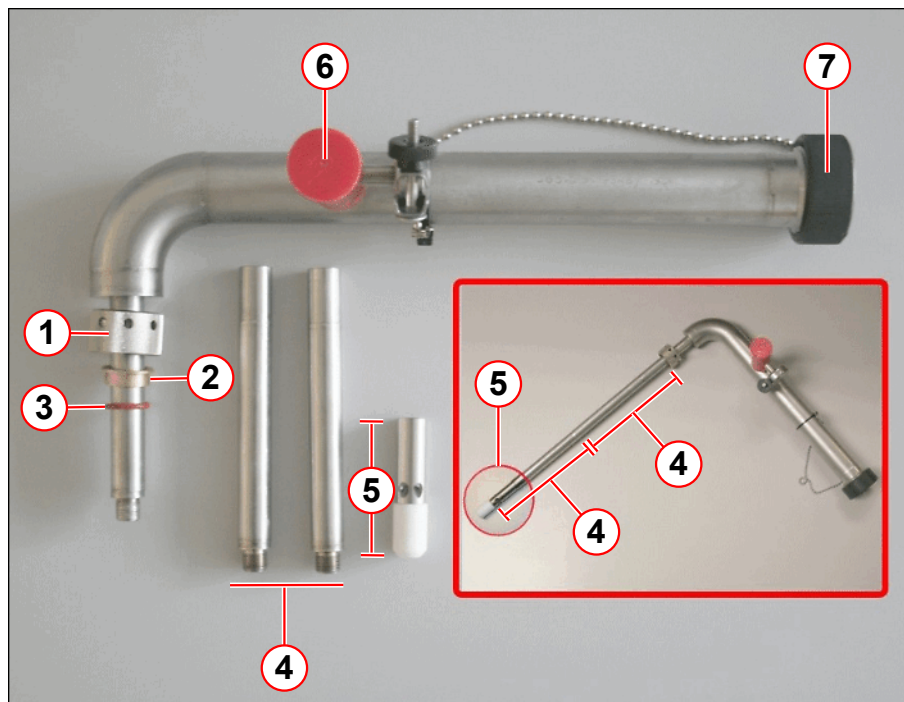


- (1) Primary ball valve
- (2) NW25 clamp and seal
- (3) Gas supply line

- Position the helium gas cylinder in a safe position.
- Fit the helium regulator to the gas bottle.
- Ensure that the **gas supply line** (pos 3) is long enough to allow the dewar to be moved close to the magnet if the magnet is at field.
- **Purge** the helium **gas supply line** of air.
- Connect it to the **primary ball valve** (1) on the dewar, using an **NW25** clamp and seal (pos 2).
- **Open** the primary ball valve and pressurize the dewar to **0.27 bar (4 psi (G))**.

1.12 Fitting the Service Syphon

Fig. 19: Magnet service syphon assembly - top filling



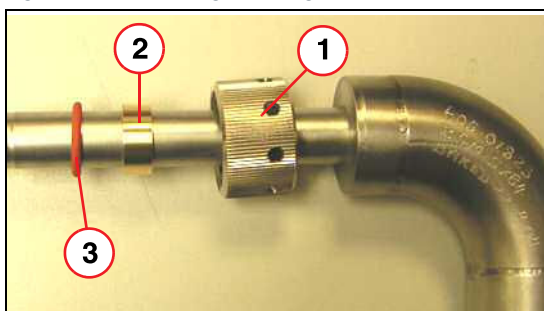
- (1) Ring nut
- (2) Brass spacer
- (3) O-ring seal
- (4) 140 mm syphon legs (2 off)
- (5) Helium fill diffuser and PTFE tip
- (6) Vacuum valve
- (7) Syphon end cap

Tab. 3 TOP filling syphon extension legs

Magnet type	140 mm leg	60 mm helium fill diffuser & PTFE tip
OR98 (Aera)	2	1
OR99 / OR103 (Skyra)		
OR103 (Spectra)		
The syphon diffuser should optimally be 50-80 mm above the syphon cone for top filling. If damaged, the PTFE diffuser tip must be replaced.		

- Check the ceiling height.
 - Fit the syphon extensions following the instructions in 'low ceiling height'. (→ Low ceiling height / Page 48)
- For normal syphon fitment:

Fig. 20: Assembling the Ring Nut Seal



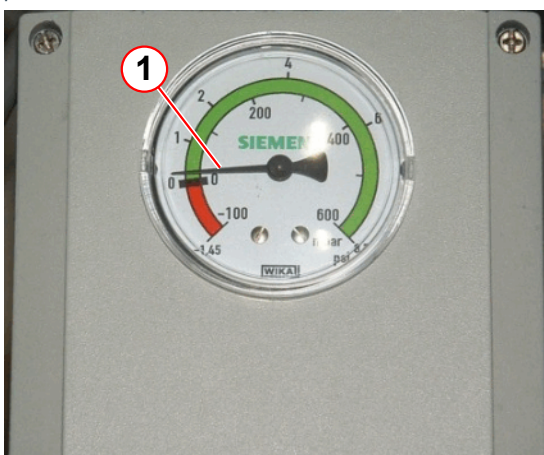
- (1) Ring nut
- (2) Brass spacer
- (3) Silicon O-ring

- Fit the ring nut, brass spacer flange, and the O-ring to the service syphon leg.

Connect the extension legs and diffuser to the syphon:

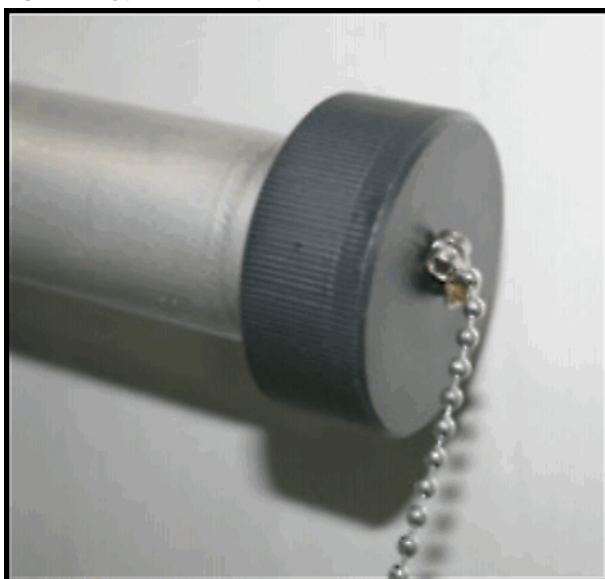
- Screw the (2 off) 140 mm legs together.
- Fit the 60 mm diffuser and PTFE tip.

Fig. 21: WIKKA Pressure switch - with a small positive pressure



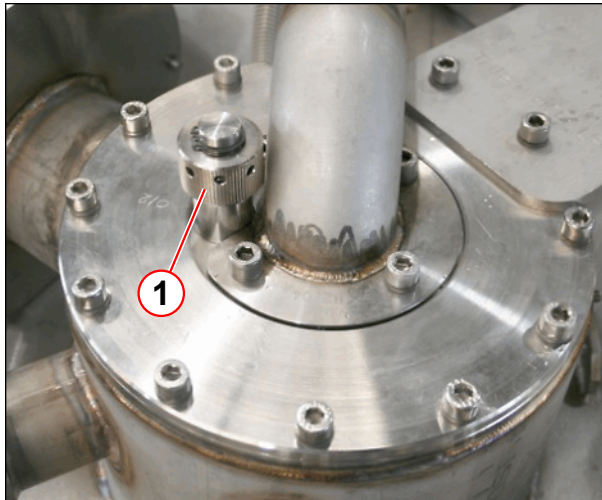
- Before fitting the syphon, check that the helium vessel has a small positive pressure.

Fig. 22: Syphon end cap



- Check that the magnet bypass valve is **closed**.
- Unscrew and remove the syphon end cap

Fig. 23: Current and Mk1 Syphon blanking plug



(1) Syphon blanking plug

- Warm and dry the syphon blanking plug using a heat gun
- Remove the syphon blanking plug using the syphon 'C' spanner.



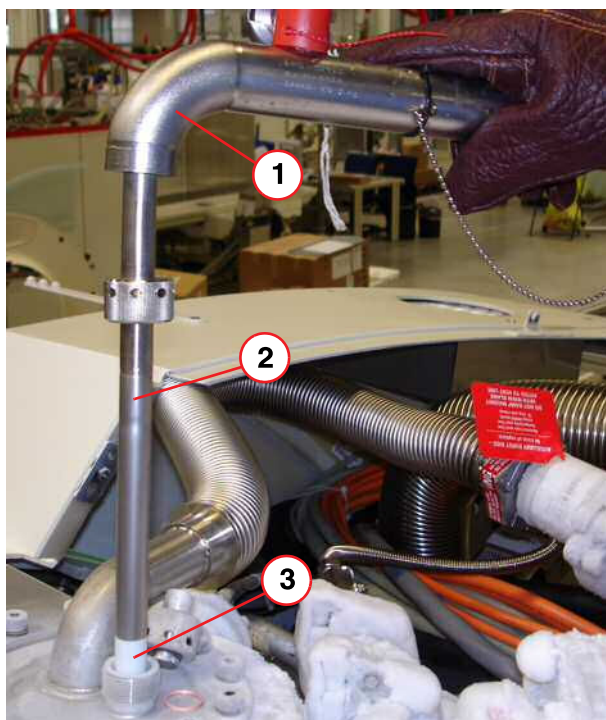
CAUTION

Risk of cold burns.

Cold gas will exhaust from the syphon port on opening.

- » Stand clear when opening.
- » Wear protective gloves and safety glasses.

Fig. 24: Helium syphon



- (1) Service Syphon
- (2) Two Extension Legs
- (3) PTFE Syphon Diffuser

- **Insert** the service syphon into the port without delay.
- Slowly lower the service syphon **fully** into the magnet.
- Tighten the syphon port ring nut by hand, then apply a half turn using the syphon C-spanner to form a **gas tight seal**.
 - Do not overtighten. The O ring seal should allow the syphon to move.
- Re-fit the syphon end cap.

1.13 Filling the Magnet -Top Fill



CAUTION

Risk of quench

Warm helium gas can cause a quench.

- » If the magnet is at field both syphons must be pre-cooled before joining together.
- » Frost or condensation on the syphon body is an indication of poor vacuum. Stop the transfer immediately - the syphon may need replacing.
- » Frost on the syphon coupling is acceptable.

The cool down of the dewar and service syphon must be performed at the same time. With less pressure in the magnet the service syphon will take longer to cool down.



For helium filling magnets at ZERO FIELD, it is not necessary to pre-cool the service and dewar syphons.

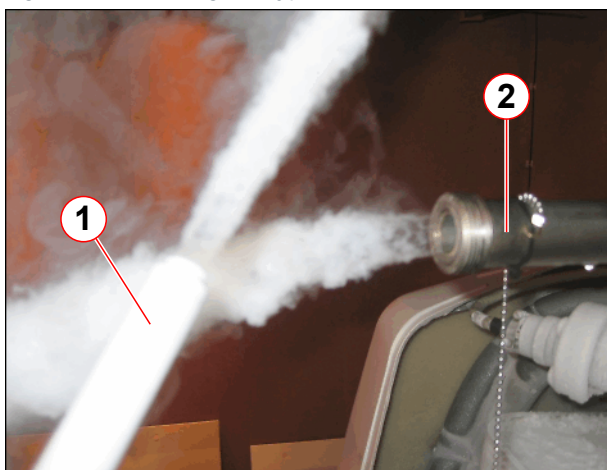
Dewar syphon cool down

- Check the dewar gas supply pressure is set at **0.27 bar (4psi (G))**.
 - Open the dewar syphon flow valve half way (2 turns).
 - Precool the syphon until a dense white **"cascading plume"** appears.
 - Close the flow valve. Wait for the service syphon to complete its cool down.

Service syphon cool down:

- Check the end cap is fitted to the service syphon.
- Check the magnet bypass valve is closed.
- If required activate the pressure heater to achieve a positive pressure of approximately **0.3 psi (G)** in the magnet.
- **Deactivate** the pressure heater, (if used to build pressure).

Fig. 25: Pre-cooling the syphons

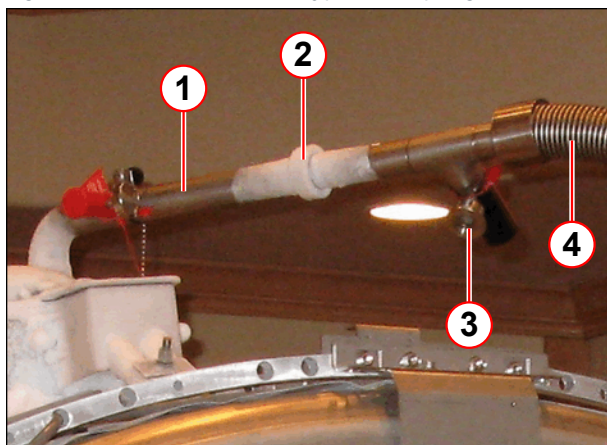


- (1) Dewar syphon
- (2) Service syphon

When there is sufficient magnet pressure to cool the service syphon:-

- Remove the syphon end cap.
- Allow the syphon to vent until **fully cold**
 - » a white “**cascading plume**” of helium gas appears.
- As the service syphon nears the point of being fully cold:-
 - » **Open** the dewar syphon flow valve half way.

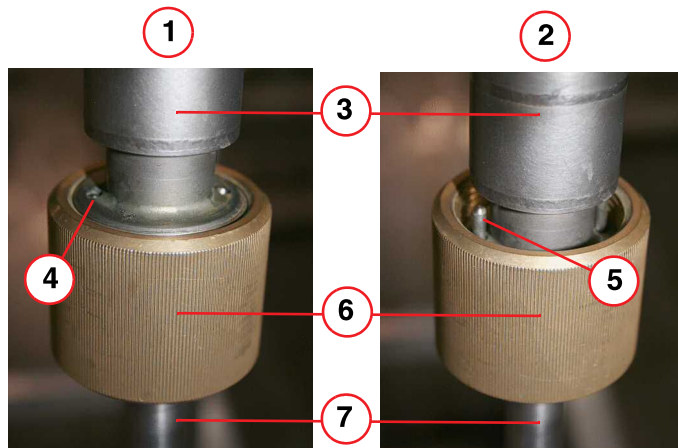
Fig. 26: Dewar and service syphon coupling



- (1) Service syphon
- (2) Lock nut
- (3) Dewar syphon vacuum valve
- (4) Dewar syphon

- When **both** syphons are giving a dense white “**cascading plume**” of helium gas, connect them together and secure the lock nut.
- **OPEN** the magnet bypass valve. This will:-
 - » Improve efficiency.
 - » Improve the filling speed.

Fig. 27: Dewar syphon flow valves - open and closed positions



- (1) FULLY OPEN position
- (2) FULLY CLOSED position
- (3) Stainless steel body
- (4) Guide pin flush with body
- (5) Guide pin
- (6) Brass nut
- (7) Syphon leg

- Fully open the Dewar syphon flow valve until the guide pins are flush with the top of the valve body.
- Check and maintain the Dewar pressure at 4 psi (G).



Frost or condensation on a syphon indicates a poor vacuum. **Stop the transfer immediately** - the syphon may need replacing.

Moderate frost on the syphon coupling (lock nut) is acceptable.

1.13.1 Dewar Transfer Pressure

The optimum pressure differential between magnet and dewar for a helium transfer is **4 psi (G)**. To calculate the required filling pressure on the helium dewars:-

- Take the magnet operating gauge pressure before depressurizing the helium vessel. and add 4 psi.
- Add **4 psi** for the required transfer pressure from the dewar.



Check manufacture's specifications for maximum dewar pressure.

1.14 Checking the Helium Fill Level

- The maximum helium level is **95%**.
- The minimum helium level for **ramping** a magnet is **40%**.
- The minimum helium level for **filling** a magnet **at field** is **40%**.
- To reduce the risk of '**alarm warning**' it is recommended that magnets are refilled to a minimum level of:
 - » Aera (OR98) and Skyra-Spectra (OR103) = **65 %**.
 - » Skyra (OR99) = **70 %**.
- Check the magnet helium levels during filling **every 5 min.**

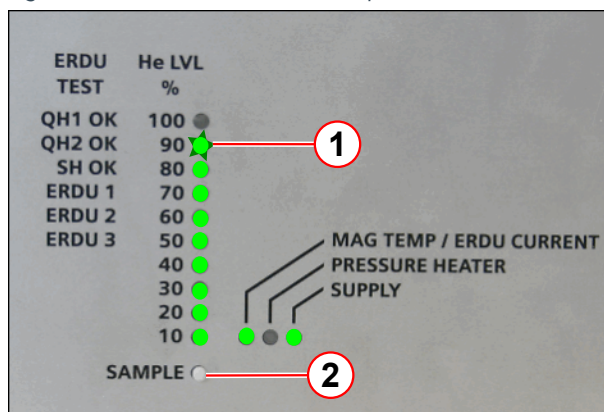
NOTICE

Helium loss

To avoid overfilling, the helium transfer must be stopped as soon as the **90 % LED** illuminates.

- » Stop the transfer at the required level.
- » Do not exceed 95%

Fig. 28: MSUP - Helium level sample



- (1) Sample switch
(2) Helium level

- Locate the **SAMPLE** switch on the MSUP lid.
- Depress the sample switch for **2 s**, and release.
- The green LEDs illuminate and display the actual liquid **helium level percentage**.
- If the 90% LED is flashing, **STOP** the transfer.

Tab. 4 MSUP Helium display

MSUP LEDs	Helium level	Comments or actions
0 - 80% ON	As displayed	Continue filling.
90% ON	As displayed	Stop transfer.
*90% FLASHING 10 - 80% ON	95% - 97%	STOP the transfer IMMEDIATELY!

MSUP LEDS	Helium level	Comments or actions
* 100% FLASHING 10% - 90% ON	97% and over	Magnet is OVERFILLED .
* The flashing LEDs return to ON after approximately 30 seconds.		

From the **service laptop**:

- Press <ESC> to display the MSUP Option screen.
- Press HT <ENTER>. Wait until the sample is completed.
- Press R <ENTER> to display the new helium value on the MSUP Process display screen.
- Maintain a pressure of **0.27 bar 4 psi (G)** on the helium dewar.
 - » Too much or too little pressure will result in inefficiency.
- Repeat the helium measurement **every 5 minutes**.
 - » An increase in level should be seen within **20 minutes**.

If not, **stop the transfer** and investigate the reason for poor transfer efficiency.



If a helium fill is showing poor fill efficiency, or no increase on the helium level meter, the transfer must be stopped and the syphon port inspected for ICE. Only trained Siemens magnet CSEs can perform de-ice procedures.



When connected to a quench line. Opening the bypass valve will help to reduce the back pressure, enabling more efficient helium transfer.

Poor fill efficiency can be caused by:

- Internal ice in the turret syphon port.
- Condensation or frost along the length of the syphon(s) indicating a poor vacuum. A replacement syphon will be required.
- Partially obstructed or blocked quench pipe.
- Partially obstructed or blocked syphon due to no diffuser fitted to the dewar syphon.

At 4 psi (G), the transfer rate is approximately **10 liters per minute**.

- Do not leave the transfer unattended. A 250 liter dewar will empty in **25 - 30 minutes**.
- Continue the helium fill until the desired helium level has been reached.

1.15 Changing Supply Dewars



CAUTION

Risk of quench.

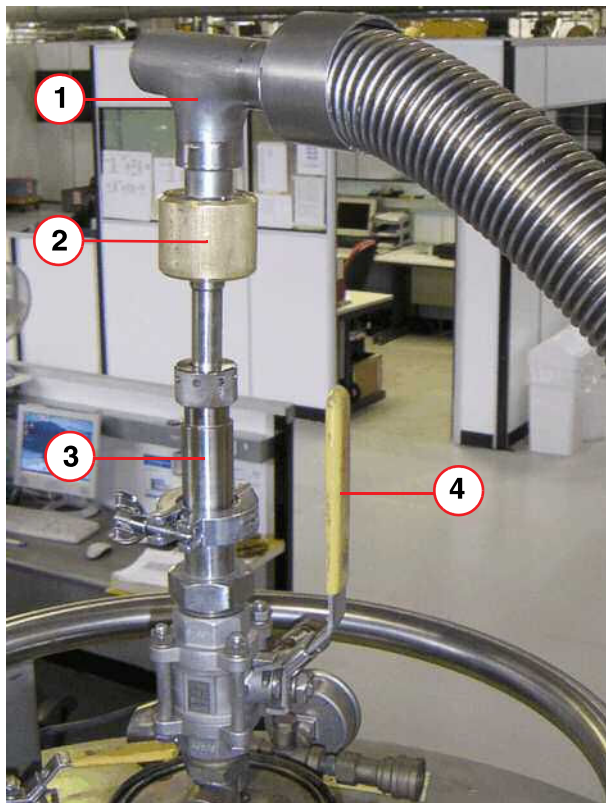
Failure to close the syphon flow valve immediately, when the magnet is at field, will result in a quench.

- » Close the syphon flow valve immediately.



Whistling of the syphon (a high pitched sound), or a fast drop in dewar pressure indicates an empty dewar.

Fig. 29: Dewar syphon fitment



- (1) Syphon
- (2) Syphon flow valve
- (3) Dewar syphon adaptor
- (4) Syphon entry valve

If the supply dewar empties before the magnet is filled, replace the supply dewar with a full one.

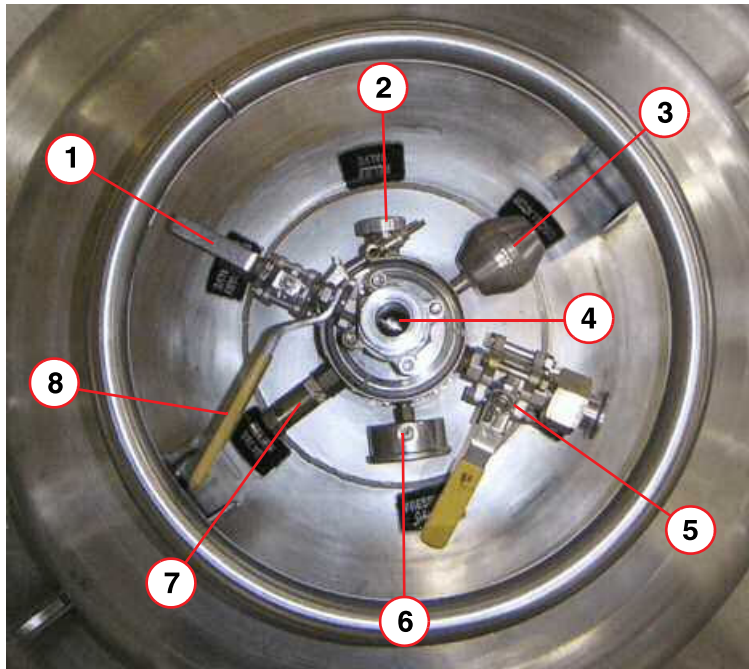
- Ensure that a full dewar is available and **depressurized**, ready to receive the syphon.
- When the dewar empties, stop the transfer **immediately** by closing the dewar syphon flow valve (pos 2).

Turn off the dewar gas supply:

- **Close** the primary ball valve and remove the gas connection.

- Vent the dewar down to atmospheric pressure in a well ventilated area.

Fig. 30: Dewar Ports (typical)



- (1) Low pressure relief valve (shown open)
- (2) Relief valve
- (3) Anti-oscillator
- (4) Syphon entry port (shown closed)
- (5) Primary ball valve (shown closed)
- (6) Pressure gauge
- (7) Relief valve
- (8) Syphon entry valve

- Using a heat gun, warm the syphon **connecting joint** to room temperature.
- Disconnect the syphons.



CAUTION

Risk of cold burns.

A pressurized flow of cold gas will exhaust from the syphons when they are disconnected.

- » Stand clear of the syphon joint when disconnecting.
- » Wear protective gloves and safety glasses.

- Remove any moisture from the end of the service syphon with paper towelling.
- Fit the service syphon end cap. (→ Fig. 22 Page 26)

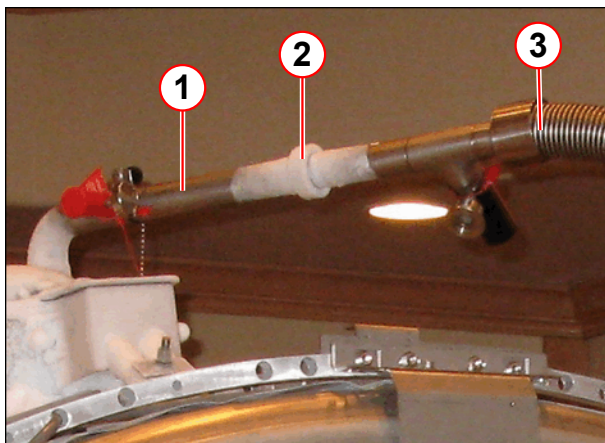


Syphon removal and fitting on top fill dewars where there is restricted ceiling height will need to be performed in a ventilated area outside the magnet room.

- Move the full dewar next to the empty dewar.

- **Open** the low pressure **relief valve** on the empty dewar to fully depressurize, (→ 1/ Fig. 30 Page 35)
- **Remove** the dewar syphon.
- **Close** the syphon entry valve on the empty dewar and **open** the low pressure relief valve.
- Remove the syphon adaptor from the empty dewar and fit onto the full dewar.
- **Close** the low pressure relief valve of the full dewar.
- **Open** the syphon flow valve **2 turns**.
- **Open** the syphon entry valve and slowly slide the syphon into the dewar. Gas will begin to flow through the syphon.
- **Purge** the gas supply line and **connect** to the full dewar.
- **Open** the dewar primary ball valve.
- Set the gas pressure to **0.27 bar – 4 psi (G)**.
- Move the dewar next to the magnet.
- **Remove** the syphon end cap of the service syphon.

Fig. 31: Dewar and service syphon coupling



- (1) Service syphon
- (2) Lock nut
- (3) Dewar syphon

- When **both** syphons are giving a dense white "**cascading plume**" of helium gas, connect them together and **secure** the lock nut.
- Continue with the transfer.

1.16 Stopping the Helium Transfer

NOTICE

Risk of quench.

Warm helium gas will cause a quench. Stop the transfer immediately when:

- » Whistling (a high pitched sound) from the syphon is audible.
- » The dewar pressure falls quickly.
- » A syphon becomes cold, frosty or wet from condensation.

- As soon as the liquid helium level reads 95% or the dewar is empty:
 - **Stop the transfer. Do not fill above 95%.**
 - Close the flow valve at the Dewar syphon.
- Indications of an empty dewar are:-
 - Whistling (high pitched sound) of the syphon.
 - A quick drop in Dewar pressure.
- The Dewar syphon flow valve must be closed immediately when the dewar is empty.
- Set the gas pressure regulator to zero.
- Close the primary ball valve on the dewar
- Remove the gas line from the dewar.

1.17 Removing the Service Syphon

- Depressurize the system helium vessel.



CAUTION

Risk of cold burns.

Cold gas will exhaust from the syphon port on opening.

- » Stand clear when opening.
- » Wear protective gloves and safety glasses.

Fig. 32: Warm up the syphon joint with heat gun.



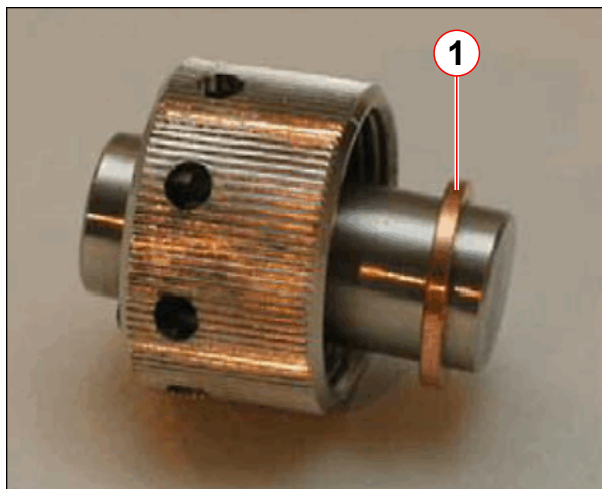
- Warm the connecting syphon joint to room temperature using a heat gun.
- Disconnect the 2 syphons.
- Immediately fit the black end cap to the service syphon.
- If necessary, warm the end of the syphon to screw the black end cap fully onto the thread.

Fig. 33: Removing the syphon



- Warm the syphon ring nut using a heat gun.
- Undo the syphon ring nut using the 'C' wrench.
- Remove the service syphon from the turret port.

Fig. 34: Syphon blanking plug



(1) Copper seal

- Immediately fit the syphon blanking plug using a new copper seal.
- Warm up the threads on the plug and carefully tighten with the syphon 'C' spanner.
 - » There might be ice on the threads - **do not use force**. Damaged threads will result in a leak.



CAUTION

Danger of air ingress.

Internal ice blockage could cause excessive pressure.

- » Warm up syphon port plug sufficiently to ensure there is no ice on the threads when tightening.

- **Close** the magnet bypass valve.

- Allow the magnet to pressurize.

1.18 Pressure Heater Activation

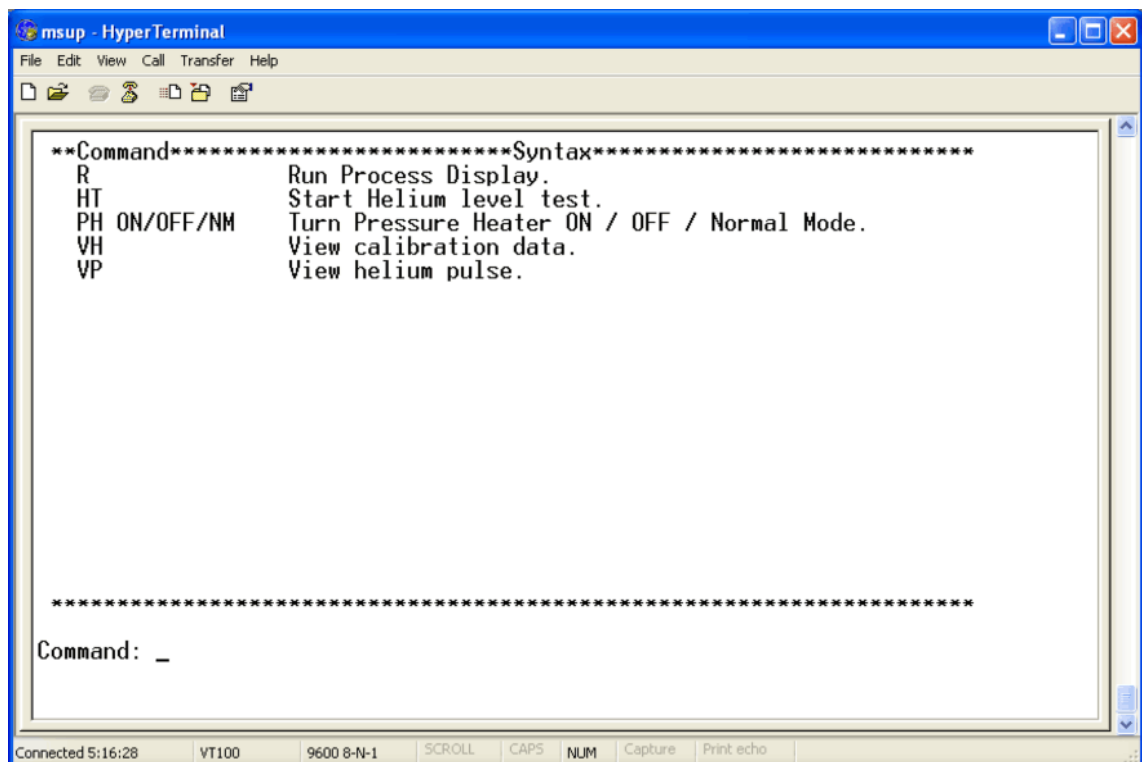
- Check that the MSUP is turned on.
- If fitted, remove the pressure service tool.



Do not use the helium vessel pressure gauge (WIKA or Mancombe) to determine the magnet pressure.

- Pressurize the helium vessel.
- Press <ESC> to display the **MSUP Option** screen (for third party companies).

Fig. 35: Option screen (Third Parties)



- Press PH ON <ENTER>.
- Press R <ENTER> to display the System Pressure on the MSUP Process display screen.
- Once the pressure reaches 15.3 / 15.5 psi (A), press <ESC> to display the MSUP Option screen.
- Press PH NM <ENTER>, press R <ENTER> to display the MSUP Option screen.
- The system will now return to normal coldhead running mode.

1.19 Final Work Steps

Helium dewar:

- Move the dewar into a well ventilated area.
- Depressurize the dewar by opening the primary ball valve.
- When at the gas pressure is at zero, close the dewar ball valve.
- Remove the dewar syphon.
- Immediately close the syphon entry valve on top of the dewar.
- Open the low-pressure dewar overpressure valve.
- Warm both syphons with the heat gun until dry, and store safely.
- Remove tools and equipment.

After transfer magnet checks:

- Check that the magnet bypass valve is **closed**.
- Check that the cold head bypass valve is **closed**.
- Check that the compressor units has switched on and is running. (Note: The 20-minute-time delay may be active).
- Check that the magnet pressure is stable at 15.3 or 15.5 psi (A). This can only be confirmed via a laptop to Msup connection. Under no circumstances is the helium vessel pressure gauge (WIKA or Mancombe) to be used as a reference point to determine that the magnet pressure is either 15.3 or 15.5 psi.

The magnet pressure stability must be monitored for 30 minutes to confirm that there are no helium gas leaks through the 16/17 psi (A) venting assembly.

- Check that the turret and venting system are warm and dry.
- Perform a leak check of all the turret and venting components.

1.20 Filling the Magnet - Bottom Fill



The bottom fill procedure must **ONLY** be performed, **Post Quench** or when the system is in **storage**. The magnet must be at **ZERO FIELD**.

Bottom filling through the magnet syphon cone is required if:-

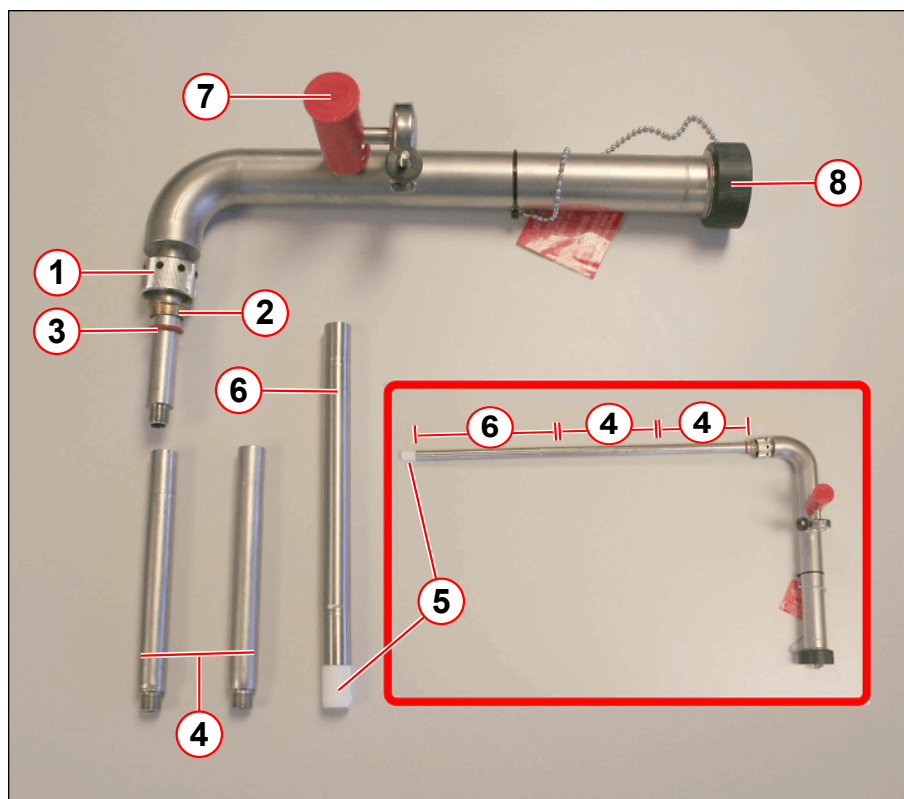
- The liquid helium level displays **zero** percent.
- Check the magnet carbon resistor (CCR) temperatures:
 - The temperatures measured must be **no warmer than** 110 K.
 - Reference measured magnet temperatures to those on the original **Magnet Acceptance Document**.



Magnet coil temperatures must be confirmed before continuing with the helium fill.

1.20.1 Assemble the service syphon

Fig. 36: Magnet service syphon assembly - bottom filling

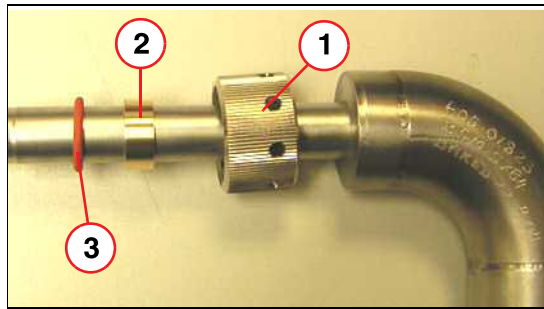


- (1) Ring nut
- (2) Brass spacer
- (3) O-ring seal
- (4) 140 mm syphon legs (2 off)
- (5) 22 mm PTFE seal (bottom fill)
- (6) 200 mm syphon leg (1 off)
- (7) Vacuum valve
- (8) Syphon end cap

Tab. 5 Bottom filling syphon extension legs

Magnet type	140 mm leg	200 mm leg	22mm PTFE seal
OR98 (Aera)	2	1	1
OR99 / OR103 (Skyra)			
OR103 (Spectra)			

Fig. 37: Assembling the Ring Nut Seal



- (1) Ring nut
- (2) Brass spacer
- (3) Silicon O-ring

- Fit the ring nut, brass spacer and the O-ring to the service syphon leg.
- Connect the extension legs and the 22 mm PTFE seal:
 - Screw the 2 x **140 mm legs** together, and connect to the syphon.
 - Fit the **1 x 200 mm leg**.
 - Fit the **22 mm PTFE seal** to the 200 mm leg.



The minimum length required to reach the syphon cone is **535 mm (+/- 5 mm)**.

1.20.2 Fit the service syphon

- Deactivate the pressure heater.
- **Depressurize** the helium vessel .
- Check the helium vessel has a **small positive pressure**.
- Warm and dry the syphon blanking plug using a heat gun to ease removal.
- Dry the syphon port.
- Remove the syphon **end cap**.
- Remove the syphon **blanking plug**.



CAUTION

Risk of cold burns.

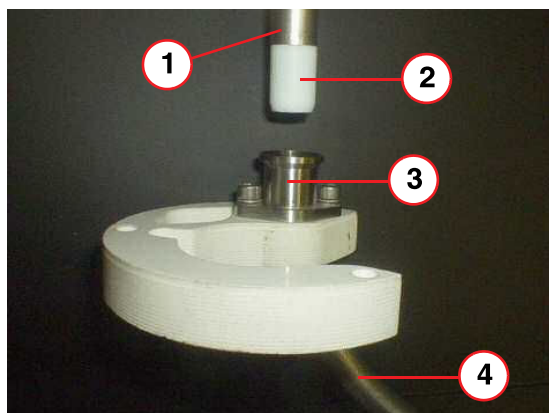
Cold gas will exhaust from the syphon port on opening.

- » **Stand clear when opening.**
- » **Wear protective gloves and safety glasses.**

- **Slowly** lower the **service syphon** fully into the neck.

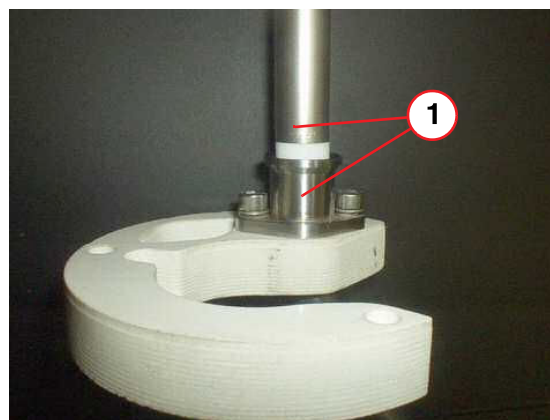
- The **syphon leg** must locate fully into the **syphon cone** - make sure it is **fully engaged**.

Fig. 38: Bottom fill syphon cone (pictorial view)



- (1) Syphon leg
- (2) PTFE Seal (Bottom Fill)
- (3) Syphon cone
- (4) Syphon tube

Fig. 39: Bottom fill - syphon leg fully engaged (pictorial view)



- (1) Syphon leg - fully engaged



Helium efficiency will be reduced if the syphon leg is not fully engaged in the syphon cone.

- Tighten the syphon port ring nut by hand, then apply a half turn, (using the appropriate spanner), to form an air-tight seal.
- Re-fit the syphon end cap.

1.20.3 Helium transfer

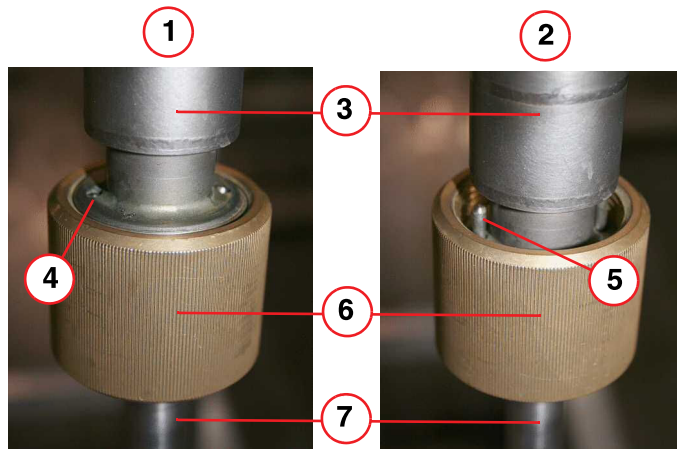


For bottom filling a magnet there is no need to pre-cool the syphons.

- Position the helium dewar close to the magnet with the dewar syphon fitted.
- **Check the dewar gas supply pressure is set at 0.27 bar (4 psi (G)).**

- **Open** the dewar syphon flow valve half way (2 full turns) to **purge** the syphon of air.

Fig. 40: Dewar syphon flow valves - open and closed positions



- (1) FULLY OPEN position
- (2) FULLY CLOSED position
- (3) Stainless steel body
- (4) Guide pin flush with body
- (5) Guide pin
- (6) Brass nut
- (7) Syphon leg

- **Close** the dewar syphon flow valve.
- Remove the service syphon end cap and connect the two syphons together. Secure the locknut.
- **Open** the dewar syphon flow valve.

- Maximize the transfer by fully opening the dewar syphon flow valve.
 - » The flow valve is **fully open** when the two guide pins at the top of the brass nut are flush with the top of the stainless steel body.
 - » From **fully closed** to **fully open**, is 4 full turns.
- Ensure the transfer dewar pressure is **4 psi (G)**.

Monitor CCR temperatures

- Monitor the **CCRs** every **15 min**, to measure the **temperature drop**.
 - » Refer to the **Acceptance documentation** (Installation Protocol) for required temperature values.
- When the CCR helium temperature is reached, continue to monitor the helium level rise.

1.20.4 Bottom fill completion

When the helium level is visibly rising on the helium level meter complete the fill following the normal **'top' fill procedures**.

- Monitor the helium fill level. (→ Checking the Helium Fill Level / Page 32)
- Stopping the helium transfer. (→ Stopping the Helium Transfer / Page 37)
- Removing the service syphon. (→ Removing the Service Syphon / Page 38)
- Changing supply dewars during transfer. (→ Changing Supply Dewars / Page 34)
- Final work steps. (→ Final Work Steps / Page 42)

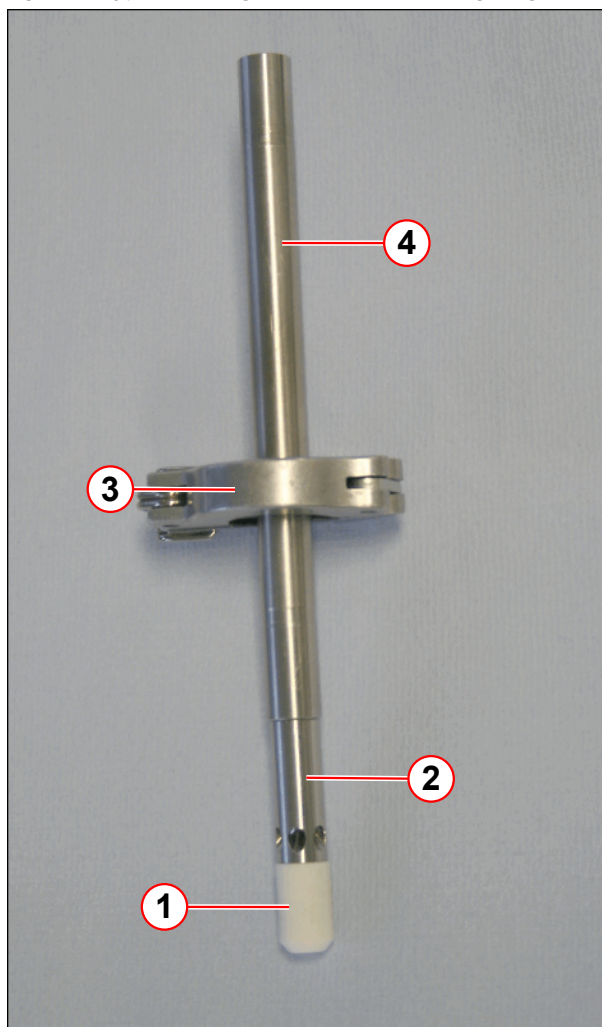
1.21 Third Party Troubleshooting

1.21.1 Low ceiling height

Where magnets have been installed into rooms with a **low ceiling height**, it can be difficult to fit the assembled service syphon. To avoid dropping the extension leg(s) down into the syphon port, the **syphon leg support clamp** must be used..

- Check the ceiling height:
 - Assemble the diffuser, bottom extension and support clamp

Fig. 41: Syphon configuration for low ceiling height



- (1) PTFE extension tip (top fill)
- (2) Helium fill diffuser tip (top fill)
- (3) Syphon leg support clamp
- (4) Extension leg

Fitting the syphon

- Remove the syphon end cap.
- Ensure the ring nut, brass spacer and O-ring are fitted correctly.
- Fit the diffuser to the extension leg.
- Fit the support clamp to the extension leg.
- Warm the syphon blanking plug to ease removal.
- Remove the syphon blanking plug using the C-spanner.
- Lower the syphon leg(s) into the neck.
- Screw on the second extension leg.
- Move the syphon clamp up the extension leg and lower extensions into the entry port, to allow room for the syphon body.
- Screw the diffuser leg to the syphon body.
- Remove the syphon clamp and slowly lower the syphon fully into the syphon entry.
- Screw the syphon port ring nut into place and tighten to make a gas-tight seal.
- Re-fit the syphon end cap.



Use the support clamp for removing the syphon.

1.21.2 Helium Volumes

Tab. 6 Magnet helium average volumes and usage

OR98			OR99 and OR103		
Probe %	Liters	Top up helium required	Probe %	Liters	Top up helium required
0	0	1500 ltrs	0	0	1350 ltrs
10	128	1350 ltrs	10	82	1200 ltrs
20	256		20	226	
30	384	1030 ltrs	30	388	925 ltrs
40	512		40	505	
50	640	750 ltrs	50	621	660 ltrs
60	768		60	733	
70	896	450 ltrs	70	851	400 ltrs
80	1024		80	970	
90	1152		90	1080	
100	1280		100	1190	
13 ltrs corresponds to 1%			12 ltrs corresponds to 1%		

1.21.3 Transfer pressure at altitude

1.21.3.1 Helium transfer with magnet bypass valve closed

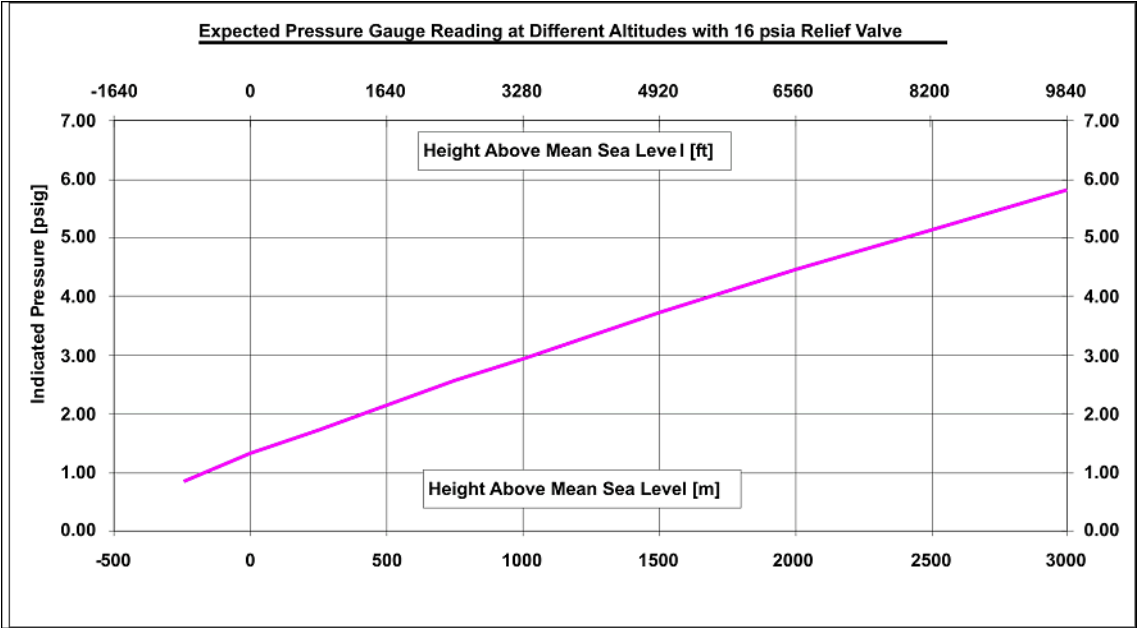
As a function of altitude, the graph(s) show the **gauge pressure** in the magnet.

The optimum pressure differential between magnet and dewar for a helium transfer is 4 psi. To calculate this:

- Determine the **magnet's altitude**
- Determine the **magnet gauge pressure** from the graphs below.

1.21.3.2 OR98 magnet - 16 psi (A) valve

Fig. 42: Effect of a change in altitude on gauge pressure (16 psi (A))



Tab. 7 Dewar transfer pressures (typical)

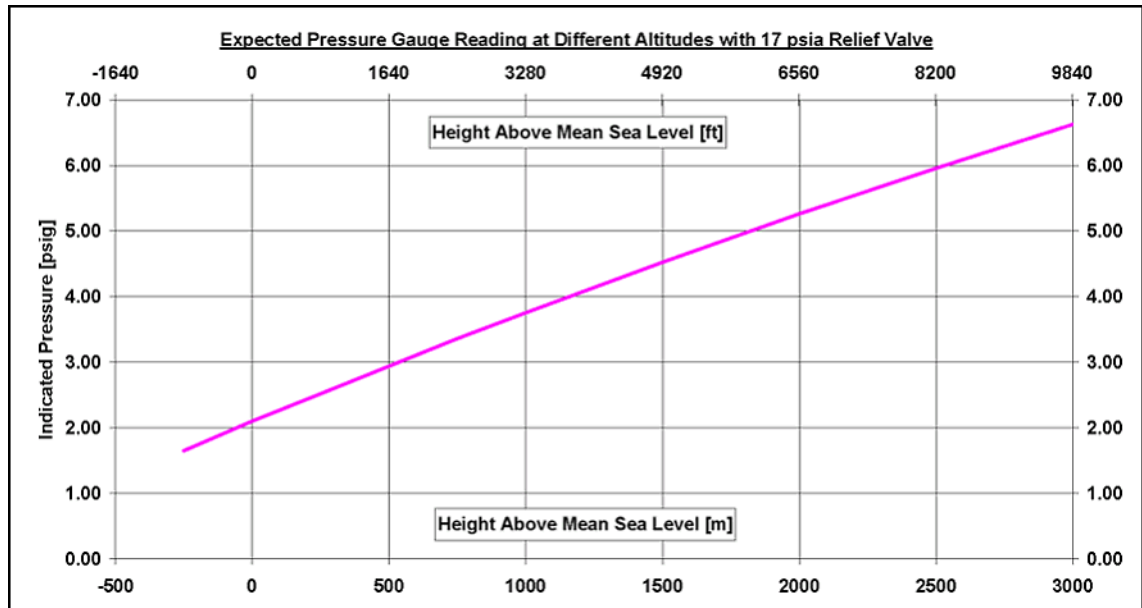
Altitude (M)	0	1000	2000	3000
Dewar pressure psi (G)	5	7.0	8.5	9.5

- Add 4 psi to this number to calculate the **required transfer pressure**
 - » **Example of 16 psi (A) vent valve assembly**
Altitude (1000m) = 3.0 psi (G) + 4 psi = 7.0 psi (G)

Maximum dewar pressure is 10.5 psi (G).

1.21.3.3 OR99/OR103 magnet 17 psi (A) valve

Fig. 43: Effect of a change in altitude on gauge pressure (17psi (A))



Tab. 8 Dewar transfer pressures (typical)

Altitude (M)	0	1000	2000	3000
Dewar pressure psi (G)	6	7.75	9.25	10.5

- Add 4 psi to this number to calculate the **required transfer pressure**
 - » Example of 17 psi (A) vent valve assembly
Altitude (1000m) = 3.75 psi (G) + 4 psi = 7.75 psi (G)



Maximum dewar pressure is 10.5 psi (G).

1.21.4 Venting leak check

After a helium fill the magnet turret and venting must be leak checked. A suitable electronic leak check device capable of measuring leaks to 1×10^{-4} range must be used.

Equipment required

- Heat gun.
- Paper towelling.
- Suitable leak detector.

The Edwards hand held leak detector is the recommended equipment. It will measure leaks greater than 1×10^{-4} mbar l/s.

Fig. 44: Edwards Leak detector (example)



Pre-requisites for leak checking

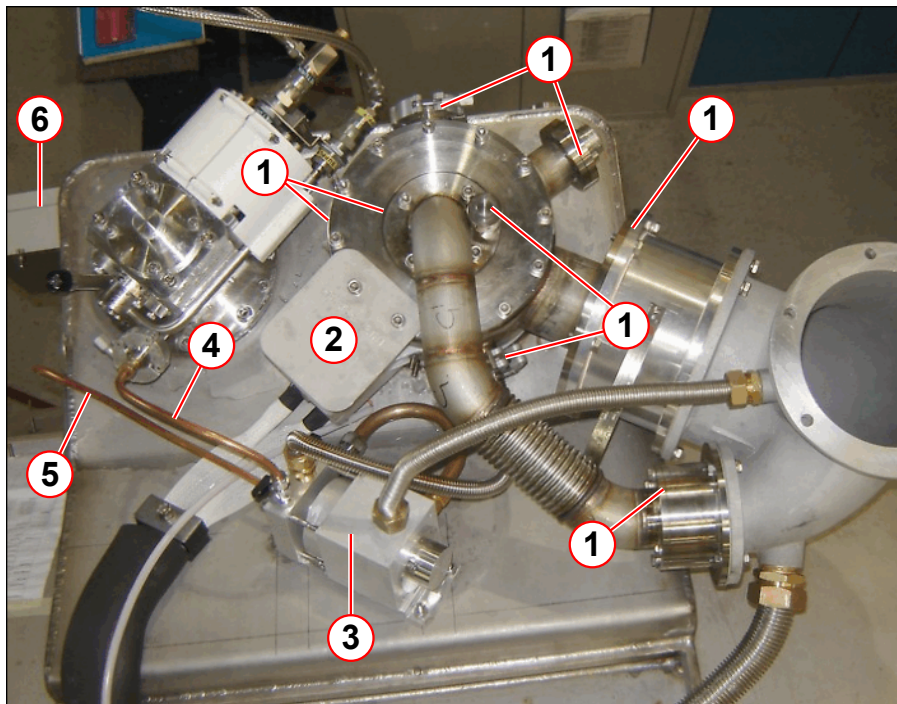
- The helium vessel must at least be at **normal operating pressure (15.3 - 15.5 psi (A))** prior to leak checking.
 - Activate the pressure heater (if required) to raise pressure.
- All joints are dry and at room temperature.
 - The syphon port plug has been re-tightened after warming around the port.



The atmospheric environment must be at least one order of magnitude lower than the smallest leak expected. After helium filling wait 30 minutes for the helium background to improve before leak checking.

- Leak check the syphon port and other seals on the turret top plate.

Fig. 45: Turret and venting leak check points - high pressure



- (1) Turret top plate, syphon port and quench valve O-rings (high pressure)
- (2) Positive FCL connector (high pressure)
- (3) Venting valve assembly
- (4) Cold head vent line
- (5) Pressure gauge vent line
- (6) Pressure gauge

- At the end of the leak check ensure that the detector is working correctly by measuring from a known helium source, (ie: helium gas bottle).
- Ensure that the system covers have been re-fitted.

Version 02

- Dokument updated.

Version 03

- Update of section (→ Pressure Heater Activation / Page 41)
- Update of section (→ Final Work Steps / Page 42)

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